

2022

Canada: Provincial Money Market Purchase Program

Lily S. Engbith

Yale School of Management

Follow this and additional works at: <https://elischolar.library.yale.edu/journal-of-financial-crises>



Part of the [Economic Policy Commons](#), [Finance and Financial Management Commons](#), [Macroeconomics Commons](#), [Policy Design, Analysis, and Evaluation Commons](#), [Policy History, Theory, and Methods Commons](#), and the [Public Administration Commons](#)

Recommended Citation

Engbith, Lily S. (2022) "Canada: Provincial Money Market Purchase Program," *Journal of Financial Crises*: Vol. 4 : Iss. 2, 1549-1568.

DOI: <https://doi.org/10.17132/2693-3179.1361>

Available at: <https://elischolar.library.yale.edu/journal-of-financial-crises/vol4/iss2/72>

This Case Study is brought to you for free and open access by the Journal of Financial Crises and EliScholar – A Digital Platform for Scholarly Publishing at Yale. For more information, please contact journalfinancialcrises@yale.edu.

READERS TAKE NOTICE:

As of April 2024, the YPFS Resource Library's site domain has changed to <https://elischolar.library.yale.edu/ypfs-financial-crisis-resource-library/>.

Please be aware that upon clicking any of the URL references to the former Resource Library domains, either <https://ypfs.som.yale.edu> or <https://ypfsresourcelibrary.blob.core.windows.net/>, in the "References/Key Program Documents" section of a case study, readers will encounter a "page not found" error.

Readers can still retrieve a given resource cited within this case study on the [new site](#) by searching [here](#) for the title cited.

Canada: Provincial Money Market Purchase Program¹

*Lily S. Engbith*²

Yale Program on Financial Stability Case Study
July 15, 2022

Abstract

In response to the COVID-19 crisis in early 2020, the Bank of Canada (BoC) enacted wide-ranging measures to support the flow of credit to individuals, firms, and municipalities. On March 24, 2020, the BoC established the Provincial Money Market Purchase Program (PMMPP) to address liquidity strains in provincial funding markets. The BoC initially committed to purchasing up to 40% of each offering of directly issued provincial money market securities with terms of 12 months or less, including provincial treasury bills and short-term promissory notes. All Canadian provinces were eligible for participation in the program. The program's utilization peaked at 7.6 billion Canadian dollars (CAD) (USD 5.7 billion) in provincial money market securities held by the BoC during the week of July 1, 2020. The announcement of the program led to a swift recovery in primary issuances and helped to tighten spreads on three-month treasury bills for Ontario and Quebec, the two most populous provinces. Citing improved conditions in short-term borrowing markets, the BoC announced the discontinuation of the PMMPP on November 9, 2020.

Keywords: asset purchases, liquidity facility, provincial securities, short-term funding, wholesale funding

¹This case study is part of the Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering market support programs in response to COVID-19. Cases are available from the *Journal of Financial Crises* at <https://elischolar.library.yale.edu/journal-of-financial-crises/>.

²Senior Research Associate, YPFS, Yale School of Management.

Overview

The spread of the COVID-19 pandemic to Canada in early March 2020 prompted the Bank of Canada (BoC) to enact a series of liquidity measures to support the flow of credit to the economy. Sharp declines in both incomes and asset values made lending riskier for banks and non-bank financial institutions, while higher margin requirements compelled institutional investors to rapidly deleverage and rebalance their portfolios. Provincial governments particularly suffered as investors became increasingly reluctant to accept new issuances of provincial debt, creating a liquidity bottleneck that prevented access to a key funding source.

In response to these liquidity strains, the BoC intervened directly by announcing the Provincial Money Market Purchase Program (PMMPP), a facility designed to promote the smooth flow of credit to provincial government funding markets. Under the PMMPP, the BoC acquired up to 40% of each offering of directly issued provincial money market securities with terms of 12 months or less, including provincial treasury bills and short-term promissory notes. All Canadian provinces were eligible for participation. Debt purchases remained as BoC assets until either each debt contract's term expired or the BoC sold the assets. The BoC intended for the PMMPP to complement the Provincial Bond Purchase Program (PBPP) and the BoC's purchases of provincial debt on a temporary basis through term repurchase operations. The BoC made its first purchases on Wednesday, March 25, 2020, and continued through November. Debt purchases made through the PMMPP would remain on the BoC's balance sheet until each debt contract's term expired or the BoC sold the debt contracts to private counterparties. Citing improved conditions in short-term funding markets, the BoC announced on July 21, 2020, that it would purchase no more than 20% of each accepted offering of directly issued provincial money market securities (BoC 2020p). It again lowered the purchasing limit to 10% on September 15, 2020, this time pointing to positive developments in the provincial borrowing markets (BoC 2020q). The BoC announced the discontinuation of the PMMPP on

Key Terms

Purpose: "To support the liquidity and efficiency of provincial government funding markets" by acquiring "provincially-issued money market securities through the primary issuance market."

Launch Dates Authorized:
March 24, 2020

Announced:
March 24, 2020

Operational Date March 25, 2020

End Date November 16, 2020

Legal Authority Bank of Canada Act,
Section 18(c)(i); March
18, 2020, Amendment

Source of Funding Bank of Canada

Administrator Bank of Canada

Overall Size Initially, 40% of each
offering of directly
issued provincial money
market securities

Purchased Assets Directly issued
provincial money
market securities

Peak Utilization CAD 7.6 billion on
July 1, 2020

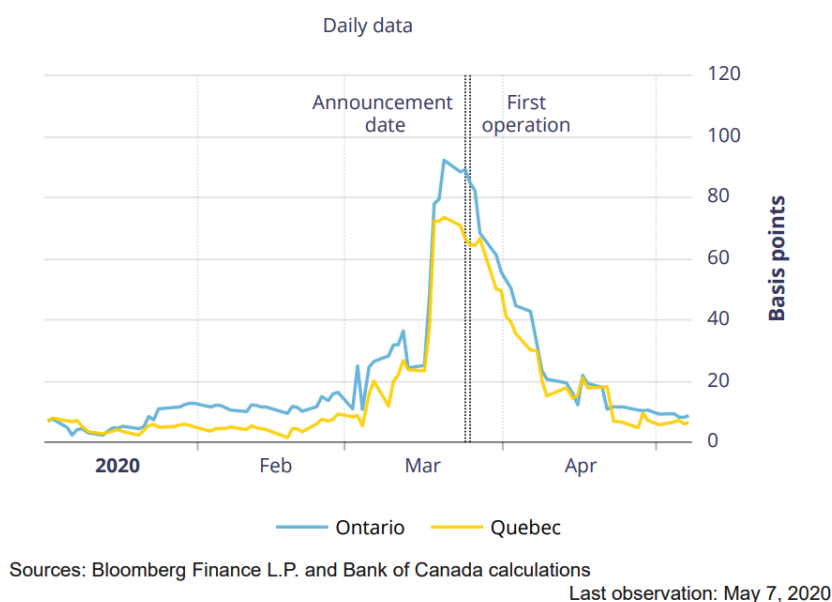
November 9, 2020. The program held its final purchasing operation on November 13, 2020, and was officially closed three days later.

BoC holdings of provincial money market securities peaked at CAD 7.6 billion (USD 5.7 billion)³ (BoC 1981–2021). On June 29, 2021, the BoC released transaction-level data for its discontinued asset purchase programs, including the PMMPP.

Summary Evaluation

The BoC responded strongly to the COVID-19 pandemic, enacting liquidity facilities and large-scale asset purchase programs to enhance liquidity in funding markets. Solovieva and Hostland called the asset purchase programs “a radical change in [the BoC’s] operations” (2020, 3). The PMMPP helped to relieve pressures in the provincial treasury bill market. According to the BoC, primary issuances recovered following the announcement on March 24, 2020, while three-month provincial treasury bill spreads for Ontario and Quebec tightened around 75 basis points (bps) between March 2020 and May 2020 (BoC 2020v, 33) (see Figure 1). However, because the effects of these programs have been studied only in aggregate, it is difficult to assess the long-term individual impact of the PMMPP on the target market (Gravelle 2021, 8; Solovieva and Hostland 2020, 3).

Figure 1: Spreads for Provincial Treasury Bills Tightened after BoC Purchases



Source: BoC 2020v, 33.

In separate statements, BoC Deputy Governors Toni Gravelle and Paul Beaudry commented that the BoC’s various emergency purchases, including those of provincial money market securities, had greatly improved access to liquidity in key funding markets, assisting with the smooth flow of credit to households and businesses (Gravelle 2021; Beaudry 2020).

³ Per Yahoo Finance, USD 1 = CAD 1.34 on March 2, 2020.

Given the improvement in financial markets over the course of 2020, Beaudry expressed confidence that the BoC's asset purchases would complement the interest rate cuts to motivate renewed investment and spending.

The BoC reported that it had incurred no credit losses under the PMMPP during the first quarter of 2020 (BoC 2020w).

Context: Canada 2019–2020	
GDP (SAAR, Nominal GDP in LCU converted to USD)	\$1.8 trillion in 2019 \$1.8 trillion in 2020
GDP per capita (SAAR, Nominal GDP in LCU converted to USD)	\$46,327 in 2019 \$43,258 in 2020
Sovereign credit rating (five-year senior debt)	Data for 2019: Moody's: Aaa S&P: AAA Fitch: AAA Data for 2020: Moody's: Aaa S&P: AAA Fitch: AA+
Size of banking system	Data not available for 2019 Data not available for 2020
Size of banking system as a percentage of GDP	Data not available for 2019 Data not available for 2020
Size of banking system as a percentage of financial system	Data not available for 2019 Data not available for 2020
Five-bank concentration of banking system	84.2% in 2019 84.5% in 2020
Foreign involvement in banking system	Data not available for 2019 Data not available for 2020
Government ownership of banking system	Data not available for 2019 Data not available for 2020
Existence of deposit insurance	Yes, in 2019 Yes, in 2020
<i>Sources: Bloomberg; FRED; World Bank Global Financial Development Database; World Bank Deposit Insurance Dataset.</i>	

Key Design Decisions

- 1. Purpose: The Bank of Canada (BoC) established the Provincial Money Market Purchase Program to address disruptions in short-term funding markets for provincial governments.**

On March 24, 2020, the BoC announced the establishment of the Provincial Money Market Purchase Program (PMMPP), a facility designed to promote the smooth flow of credit to provincial government funding markets. The PMMPP complemented the Provincial Bond Purchase Program and special term repurchase operations intended to support provincial governments.

- 2. Part of a Package: The BoC introduced a variety of asset purchase facilities to mitigate pandemic-related liquidity strains.**

Considering the global volatility, abrupt drop in oil prices, and potential for “permanent damage to the structure of the economy” related to the outbreak of COVID-19 in Canada, the central bank instituted a large package of support measures equal to about 12% of GDP (BoC 2020i). Large-scale asset purchases of Canada Mortgage Bonds, bankers’ acceptances, commercial paper, corporate bonds, and provincial and federal government debt allowed the BoC to support market functioning, while an expansion of term repo facilities addressed credit availability for businesses and municipalities. Additionally, on March 27, 2020, the BoC lowered the overnight target rate to its effective lower bound of 25 bps, down from 75 bps just two weeks prior.

- 3. Legal Authority: The BoC gave itself the ability to buy and sell provincial securities outright, as allowed under Section 18(c) of the Bank of Canada Act.**

Subsection 18(c) of the Act allows the BoC to “buy and sell securities issued or guaranteed by Canada or any province” (Bank of Canada Act 1985).

- 4. Governance: Aside from standing internal audit powers authorized under the Bank of Canada Act, there were seemingly no external audits or oversight of the PMMPP’s operations.**

Section 28 of the Bank of Canada Act enables the Minister of Finance to appoint auditors of the BoC and to require that those auditors report on bank affairs to the Minister “from time to time” (Bank of Canada Act 1985).

- 5. Administration: The BoC was the sole administrator of the PMMPP.**

The BoC purchased securities directly from issuing provincial governments. Because of the existing legal provisions in the Bank of Canada Act, the BoC was able to implement the program immediately. The BoC managed the PMMPP internally and did not contract other government agencies or private entities in the establishment or operation of the PMMPP.

6. Communication: The BoC continuously reassured the public that the actions it undertook in response to the COVID-19 pandemic were intended to support market functioning and establish a strong foundation for economic recovery.

In announcing its intent to establish numerous programs to support the financial system, the BoC emphasized the importance of returning the economy to a state of normalcy. The Governing Council of the BoC also committed to adjusting its measures and taking further action “as required” in response to the evolving situation (BoC 2020i, 1). Furthermore, in his opening statement before the House Commons Standing Committee on Finance, Governor Stephen Poloz emphasized that the BoC “[stood] ready to augment the scale of any of its programs should market conditions warrant it” (Poloz 2020, 1).

7. Disclosure: Beyond requirements specified by Section 29 of the Bank of Canada Act, the BoC published weekly aggregate holdings of provincial money market securities acquired through the program.

The BoC reported aggregate holdings of provincial money market securities acquired through the PMMPP every Friday on the bank’s weekly balance sheet; the first results were published on March 25, 2020.

This reporting appears to go beyond what was required in Section 29 of the Bank of Canada Act. Subsection 29(1) required the BoC to post aggregate information about its assets and liabilities on its website each Friday. Subsection 29(2) required the BoC to report monthly on investments and securities issued or guaranteed by the government of Canada (Bank of Canada Act 1985).

The Governor of the Bank or the Minister of Finance could compel the BoC to submit a summary or report along with these financial statements for publication in the *Canada Gazette*. The Minister of Finance would then present such documents to Parliament within three weeks (Bank of Canada Act 1985).

The BoC upheld its stated commitment to transparency in releasing transaction-level data for its discontinued asset purchase programs introduced during the COVID-19 pandemic. Although not required to explicitly by law, the BoC announced upon termination of the programs that it would disclose the names of individual issuers and amounts of provincial money market securities purchased over the course of the program (BoC 2021c). On June 29, 2021, the BoC published on its website transaction-level data for its emergency asset purchase programs, including the PMMPP.

8. Use of SPV: The BoC did not utilize a special purpose vehicle (SPV) to purchase money market securities from program participants.

The BoC purchased securities directly from eligible issuers.

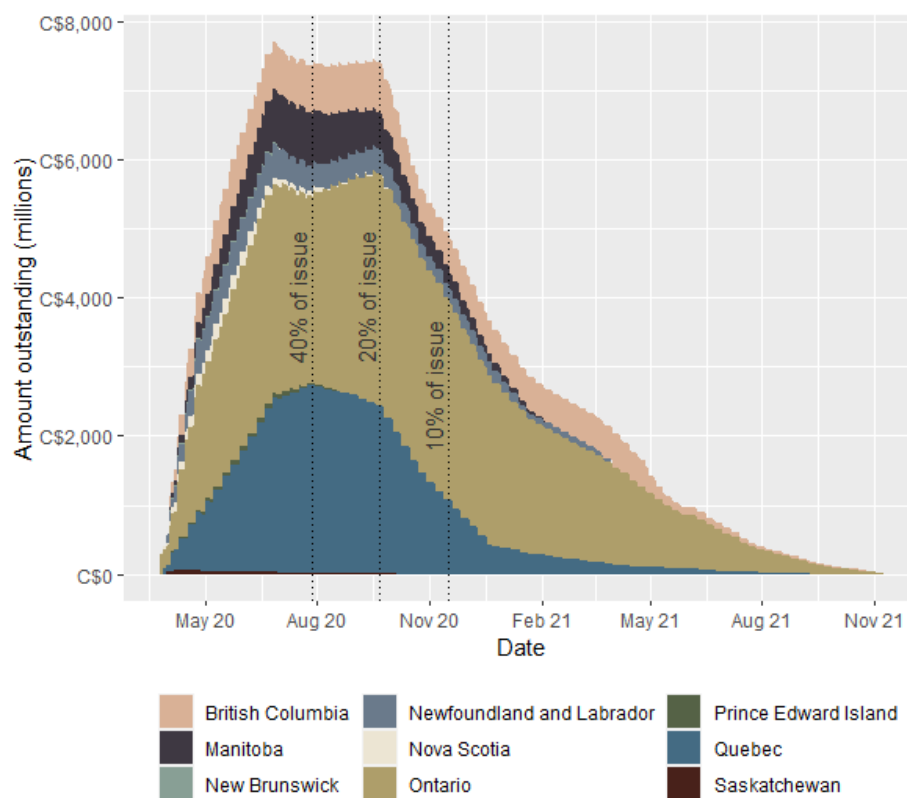
9. Size: The BoC initially committed to purchasing up to 40% of each offering of directly issued provincial money market securities.

The size of the facility was effectively limited by the aggregate amount of money market securities issued by provinces.

The BoC could purchase up to 40% of one specific bill or across all bills, at times purchasing the entire 40% of each offering; the decision depended on the maturity of the bill and the underlying issuer. Citing improved conditions in short-term funding markets, the BoC announced on July 21, 2020, that it would reduce the purchase limit from 40% to 20% (BoC 2020p). It again reduced the threshold from 20% to 10%, stating in a press release on September 15, 2020, that the short-term provincial borrowing market had shown “significant” signs of recovery (BoC 2020q, 1).

As seen in Figure 2, program utilization peaked at CAD 7.6 billion in provincial money market securities held by the BoC the week of July 1, 2020.

Figure 2: BoC Provincial Money Market Securities Holdings



Source: BoC 1981–2021.

10. Source of Funding: The BoC funded the PMMPP with settlement balances and received an indemnification guarantee from the government for any losses incurred under the program.

In an explanatory note on the BoC's use of reserves for emergency purposes, Deputy Governor Paul Beaudry analogized settlement balances—deposits to the facility's account with the central bank—with loans from institutions to the central bank. The BoC credited issuers upon purchase with a deposit of settlement balances, paying interest at the policy rate of 25 bps (Beaudry 2020). This policy derived from the BoC's existing "Statement of Policy Governing the Acquisition and Management of Financial Assets for the Bank of Canada's Balance Sheet."

The BoC also entered into independent financial guarantee contracts with the government of Canada that indemnified the central bank in the case that it incurred credit losses on assets purchased under the PMMPP. In accordance with International Accounting Standard 37, these agreements stipulated that assets would be recognized only if an issuer were in default and the BoC had filed a claim of reimbursement with the government (BoC 2021a). The BoC reported that it had incurred no credit losses under the PMMPP during the first quarter of 2020 (BoC 2020w).

According to the BoC, the purchases of bankers' acceptances and provincial money market securities together resulted in a "significant" rise in deposits (BoC 2020w, 4). Compared with other programs enacted in response to COVID-19, however, the PMMPP made only a minor contribution to the substantial increase in the BoC's balance sheet. Purchasing operations conducted under the PMMPP resulted in a balance of CAD 569.7 million (at amortized cost) as of March 31, 2020 (BoC 2020w). Figures 3 and 4 illustrate the relatively small impact of provincial money market securities on the BoC's total asset holdings.

Figure 3: Summary of Assets Held by the BoC as of September 30, 2020**Assets****Summary of assets**

(in millions of Canadian dollars)

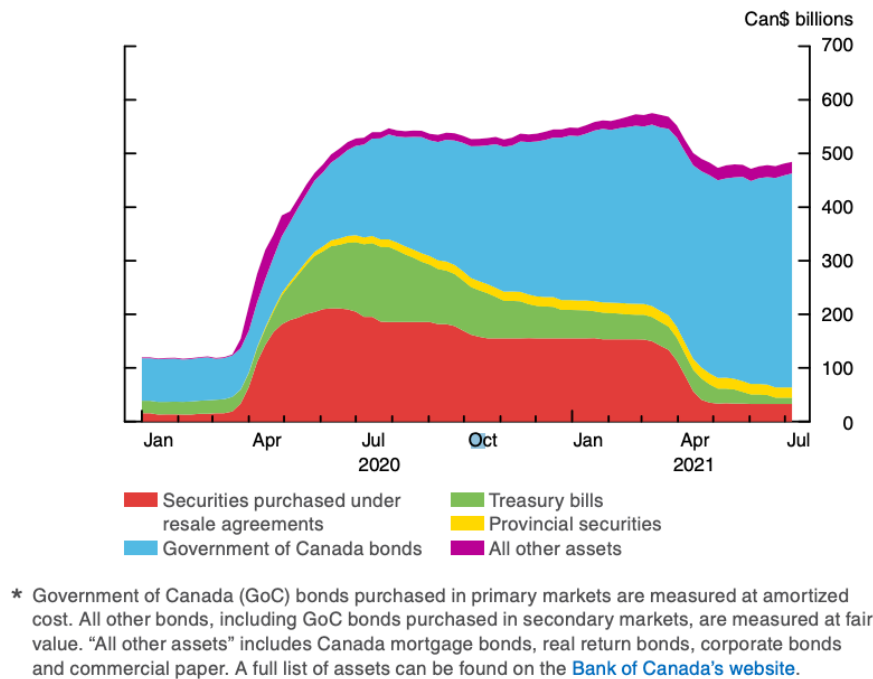
As at	September 30, 2020	December 31, 2019	Variance	
			\$	%
Cash and foreign deposits	6.8	6.4	0.4	6
Loans and receivables				
Securities purchased under resale agreements	169,950.8	15,516.5	154,434.3	995
Advances to members of Payments Canada	150.3	-	150.3	-
Other receivables	4.9	5.4	(0.5)	(9)
	170,106.0	15,521.9	154,584.1	996
Investments				
Government of Canada treasury bills	93,268.6	23,367.4	69,901.2	299
Government of Canada bonds—carried at amortized cost	94,196.1	79,030.5	15,165.6	19
Government of Canada bonds—carried at fair value through profit and loss	147,493.3	-	147,493.3	-
Canada Mortgage Bonds	9,133.3	510.7	8,622.6	1,688
Other bonds*	9,537.7	-	9,537.7	-
Securities lent or sold under repurchase agreements	1,032.7	-	1,032.7	-
Other securities†	7,036.1	-	7,036.1	-
Shares in the Bank for International Settlements	505.5	438.3	67.2	15
	362,203.3	103,346.9	258,856.4	250
Capital assets‡	691.9	700.9	(9.0)	(1)
Other assets	32.9	66.7	(33.8)	(51)
Total assets	533,040.9	119,642.8	413,398.1	346

* Includes Provincial bonds and Corporate bonds

† Includes Provincial money market securities

‡ Includes Property and equipment, Intangible assets and Right-of-use leased assets

Source: BoC 2020aa, 4.

Figure 4: The BoC's Total Assets as of July 7, 2021—Weekly Data

Source: BoC 2021g, 24.

11. Eligible Institutions: All Canadian provinces were eligible for participation.

There do not appear to have been any further restrictions on eligibility.

12. Auction or Standing Facility: Using the PMMPP as a standing facility, the BoC would directly purchase provincial money market securities at the issuers' election.

The BoC purchased eligible assets upon each accepted offering.

13. Loan or Purchase: The BoC purchased provincial money market securities issued on the primary market.

The BoC carried out purchases following direct offerings of provincial money market securities.

14. Eligible Assets: The PMMPP purchased directly issued provincial money market securities with terms of 12 months or less, including treasury bills and short-term promissory notes.

Eligible assets included treasury bills and short-term promissory notes directly issued by any Canadian province.

15. Purchase Price: The BoC purchased provincial money market securities at fair value according to its fair value hierarchy.

The BoC purchased eligible assets through the primary market using “market prices derived from observable inputs,” as specified by Level 2 of the fair value hierarchy (BoC 2020w, 26–27).⁴ These prices are derived either directly (e.g., from prices for similar instruments or inactive markets) or indirectly (e.g., from interest rates or credit spreads) (BoC 2020w).

16. Haircuts: This Key Design Decision does not apply to asset purchase programs.

Haircuts apply only to programs that involve lending to counterparties.

17. Interest Rates: This Key Design Decision does not apply to asset purchase programs.

Interest rates apply only to programs that involve lending to counterparties.

18. Fees: There were no fees associated with the PMMPP.

Sources consulted did not specify any participation fees related to use of the program.

19. Term/Repayment: This Key Design Decision does not apply to asset purchase programs.

This refers to the loan maturity and applies only to programs that involve lending to counterparties.

20. Other Restrictions on Eligible Participants: There were no restrictions associated with participation in the program.

Sources consulted do not indicate the imposition of restrictions on PMMPP participants.

⁴ The BoC designed the fair value hierarchy to reflect the “significance of the inputs” used to determine the following measurements:

“Level 1: Unadjusted quoted prices in active markets for identical assets or liabilities, which represent actual and regularly occurring arm’s-length market transactions.

“Level 2: Inputs other than quoted prices included in Level 1, which are observable for the assets or liabilities either directly (e.g., prices for similar instruments, prices from inactive markets) or indirectly (e.g., interest rates, credit spreads).

“Level 3: Unobservable inputs for the assets or liabilities that are not based on observable market data due to inactive markets (e.g., market participant assumptions)” (BoC 2020w, 26).

Fair value measurements classified financial instruments at the lowest level of the hierarchy “for which a significant input has been considered” (BoC 2020w, 26).

21. Regulatory Relief: The BoC did not implement any regulatory changes to assist with the operation of—or participation in—the program.

Sources consulted do not indicate the existence of regulatory concessions.

22. International Coordination: The BoC did not coordinate with other jurisdictions when designing or operating the PMMPP.

Sources consulted do not indicate that the BoC consulted with international parties.

23. Duration: The PMMPP carried out its final purchasing operation on November 13, 2020.

According to documents consulted, the BoC did not initially establish an end date for the PMMPP. It announced in early November that it would terminate the PMMPP on November 16, 2020, following the program's last operation on November 13, 2020. The BoC pointed to "continued improvement in the functioning of short-term funding markets and financial markets more generally" in making its decision (BoC 2020r, 1).

References and Key Program Documents

Summary of Program

(BoC 2020a) Bank of Canada (BoC). 2020. "Provincial Money Market Purchase Program." *BoC webpage describing operational terms and conditions of the PMMPP*.
<https://ypfs.som.yale.edu/node/18376>

Legal/Regulatory Guidance

(Bank of Canada Act 1985) 1985. c. B-2 Revised Statutes of Canada, 1985.
Canadian law establishing the Bank of Canada and outlining its purpose and powers.
<https://ypfs.som.yale.edu/library/bank-canada-act>

(BoC 2015) Bank of Canada (BoC). October 3, 2015. "Amendment to Policy for Buying and Selling Securities." *Canada Gazette*, Part I 149 (40).
Establishment of policy outlining how the BoC will use its 18(g)(i) authority to conduct repurchases of private-sector securities.
<https://ypfs.som.yale.edu/node/18401>

(BoC 2020b) Bank of Canada (BoC). March 13, 2020. "Amendment to Policy for Buying and Selling Securities." *Canada Gazette*, Part I 154 (1): 1–4.
BoC policy amending 2015 policy to allow for outright purchases and sales of private-sector securities in the service of monetary policy for financial stability.
<https://ypfs.som.yale.edu/node/18402>

(BoC 2020c) Bank of Canada (BoC). March 18, 2020. "Amendment to Policy for Buying and Selling Securities." *Canada Gazette*, Part I 154 (2): 1–4.
BoC policy amending 2015 policy to allow for outright purchases and sales of private-sector securities in the service of monetary policy for financial stability.
<https://ypfs.som.yale.edu/node/18300>

(BoC 2021a) Bank of Canada (BoC). May 1, 2021. "Government Notices." *Canada Gazette* Part I: Vol. 155 (18).
Government notice regarding regulatory changes made by Bank of Canada, as of May 2021.
<https://ypfs.som.yale.edu/node/18281>

(BoC 2021b) Bank of Canada/Central Bank of Canada/La Banque du Canada (BoC). September 1, 2021. "UPDATED: Statement of Policy Governing the Acquisition and Management of Financial Assets for the Bank of Canada's Balance Sheet (Effective: September 1, 2021)." Bank of Canada/Central Bank of Canada/La Banque du Canada. Yale Program on Financial Stability Resource Library.
Official BoC statement regarding the funding, governance, administration, risk management, and other operational details of the Bank's asset purchase programs.
<https://ypfs.som.yale.edu/node/18391>

Press Releases/Announcements

(Beaudry 2020) Beaudry, Paul. August 2020. "Our COVID-19 Response: Large-Scale Asset Purchases." Bank of Canada. Yale Program on Financial Stability Resource Library.

Public-facing summary describing the BoC's large-scale asset purchase program operations.
<https://ypfs.som.yale.edu/node/18298>

(BoC 2020d) Bank of Canada (BoC). March 13, 2020. "Bank of Canada Lowers Overnight Rate Target to $\frac{3}{4}$ Percent." Bank of Canada. Yale Program on Financial Stability Resource Library.

Press release announcing the BoC's decision to lower the overnight rate target to $\frac{3}{4}$ percent.
<https://ypfs.som.yale.edu/node/18494>

(BoC 2020e) Bank of Canada (BoC). March 20, 2020. "Bank of Canada Announces Additional Measures to Support Market Functioning." Bank of Canada.

BoC notice announcing new programs to support Canadian markets.
<https://ypfs.som.yale.edu/node/18197>

(BoC 2020f) Bank of Canada (BoC). March 24, 2020. "Bank of Canada Announces New Program to Support Provincial Funding Markets." Bank of Canada. Yale Program on Financial Stability Resource Library.

BoC press release announcing the PMMPP to support market liquidity for provincial short-term borrowing.
<https://ypfs.som.yale.edu/node/18377>

(BoC 2020g) Bank of Canada (BoC). March 25, 2020. "Results of Today's Provincial Money Market Purchase Program." Bank of Canada. Yale Program on Financial Stability Resource Library.

BoC press release announcing the daily results of purchases made under the PMMPP.
<https://ypfs.som.yale.edu/node/18378>

(BoC 2020h) Bank of Canada (BoC). March 26, 2020. "Results of Today's Provincial Money Market Purchase Program." Bank of Canada. Yale Program on Financial Stability Resource Library.

BoC press release announcing the daily results of purchases made under the PMMPP.
<https://ypfs.som.yale.edu/node/18379>

(BoC 2020i) Bank of Canada (BoC). March 27, 2020. "Bank of Canada Lowers Overnight Rate Target to $\frac{1}{4}$ Percent." Bank of Canada. Yale Program on Financial Stability Resource Library.

Bank of Canada notice announcing an interest rate cut in response to COVID-19.
<https://ypfs.som.yale.edu/node/18260>

(BoC 2020j) Bank of Canada (BoC). March 30, 2020. "Results of Today's Provincial Money Market Purchase Program." Bank of Canada. Yale Program on Financial Stability Resource Library.

BoC press release announcing the daily results of purchases made under the PMMPP.
<https://ypfs.som.yale.edu/node/18380>

(BoC 2020k) Bank of Canada (BoC). March 31, 2020. "Results of Today's Provincial Money Market Purchase Program." Bank of Canada. Yale Program on Financial Stability Resource Library.

BoC press release announcing the daily results of purchases made under the PMMPP.
<https://ypfs.som.yale.edu/node/18381>

(BoC 2020l) Bank of Canada (BoC). April 2, 2020. "Results of Today's Provincial Money Market Purchase Program." Bank of Canada. Yale Program on Financial Stability Resource Library.

BoC press release announcing the daily results of purchases made under the PMMPP.
<https://ypfs.som.yale.edu/node/18383>

(BoC 2020m) Bank of Canada (BoC). April 6, 2020. "Results of Today's Provincial Money Market Purchase Program." Bank of Canada. Yale Program on Financial Stability Resource Library.

BoC press release announcing the daily results of purchases made under the PMMPP.
<https://ypfs.som.yale.edu/node/18384>

(BoC 2020n) Bank of Canada (BoC). April 8, 2020. "Results of Today's Provincial Money Market Purchase Program." Bank of Canada. Yale Program on Financial Stability Resource Library.

BoC press release announcing the daily results of purchases made under the PMMPP.
<https://ypfs.som.yale.edu/node/18385>

(BoC 2020o) Bank of Canada (BoC). April 9, 2020. "Results of Today's Provincial Money Market Purchase Program." Bank of Canada. Yale Program on Financial Stability Resource Library.

BoC press release announcing the daily results of purchases made under the PMMPP.
<https://ypfs.som.yale.edu/node/18386>

(BoC 2020p) Bank of Canada (BoC). July 21, 2020. "Bank of Canada Announces Changes to the Provincial Money Market Purchase (PMMP) Program." Bank of Canada. Yale Program on Financial Stability Resource Library.

BoC press release announcing that the Bank would purchase up to 20% of each accepted offering.
<https://ypfs.som.yale.edu/node/18387>

(BoC 2020q) Bank of Canada (BoC). September 15, 2020. "Bank of Canada Announces Changes to the Provincial Money Market Purchase (PMMP) Program." Bank of Canada/Central Bank of Canada/La Banque du Canada. Yale Program on Financial Stability Resource Library.

BoC press release announcing that the Bank would purchase up to 10% of each accepted offering.
<https://ypfs.som.yale.edu/node/18388>

(BoC 2020r) Bank of Canada (BoC). November 9, 2020. "Bank of Canada Announces Amendments to Targeted Facilities and Programs That Support Liquidity and Short-Term Funding Markets." Bank of Canada. Yale Program on Financial Stability Resource Library. *BoC press release announcing the discontinuation of the PMMPP on November 16, 2020.* <https://ypfs.som.yale.edu/node/18389>

(BoC 2021c) Bank of Canada (BoC). March 23, 2021. "Bank of Canada Announces the Discontinuation of Market Functioning Programs Introduced during COVID-19." *Press release announcing discontinuation of PBPP, as well as other COVID-19 programs.* <https://ypfs.som.yale.edu/node/18159>

(BoC 2021d) Bank of Canada (BoC). June 29, 2021. "Bank of Canada Publishes Transaction-Level Data for Its Discontinued Asset Purchase Programs." Bank of Canada/Central Bank of Canada/La Banque du Canada. Yale Program on Financial Stability Resource Library. *Press release announcing publication of transaction-level data for asset purchase programs.* <https://ypfs.som.yale.edu/node/18261>

(Poloz 2020) Poloz, Stephen S. April 16, 2020. "Opening Statement before the House of Commons Standing Committee on Finance." Ottawa, April 16. *Opening statement delivered to Canada's House of Commons Standing Committee on Finance regarding the BoC's monetary policy and liquidity actions.* <https://ypfs.som.yale.edu/node/18295>

Media Stories

(Argitis and Hertzberg 2020) Argitis, Thophilos, and Erik Hertzberg. March 24, 2020. "Bank of Canada Extends Asset Buying to Provincial Money Markets." Bloomberg L.P. Yale Program on Financial Stability Resource Library. *Media story reporting on the establishment of the BoC's support for provincial governments.* <https://ypfs.som.yale.edu/node/18442>

(Evans 2020) Evans, Pete. March 27, 2020. "Bank of Canada Makes Another Emergency Cut to Interest Rate." CBC News. Yale Program on Financial Stability Resource Library. *News article reporting BoC's second COVID-related rate cut in two weeks.* <https://ypfs.som.yale.edu/node/18246>

Key Academic Papers

(Fontaine, Selody, and Wilkins 2009) Fontaine, Jean-Sébastien, Jack Selody, and Carolyn Wilkins. December 2009. "Improving the Resilience of Core Funding Markets." Financial System Review. *Bank of Canada research report analyzing the country's core funding markets and policies to improve stability.* <https://ypfs.som.yale.edu/node/18237>

Reports/Assessments

(Bédard-Pagé, Bolduc, Demers, Dion, Pandey, Berger-Soucy, and Walton 2021) Bédard-Pagé, Guillaume, Daniel Bolduc, Annick Demers, Jean-Philippe Dion, Manu Pandey, Léanne Berger-Soucy, and Adrian Walton. May 28, 2021. "COVID-19 Crisis: Liquidity Management at Canada's Largest Public Pension Funds." Staff Analytical Note 2021-11. Yale Program on Financial Stability Resource Library.

BoC Staff Analytical Note regarding Canada's pension funds and their liquidity management during 2020.

<https://ypfs.som.yale.edu/node/18256>

(BoC 1981-2021) Bank of Canada (BoC). 1981-2021. "Bank of Canada Assets and Liabilities: Weekly (Formerly B2)." Bank of Canada. Yale Program on Financial Stability Resource Library.

Weekly Wednesday statement disclosing assets and liabilities as required by the Bank of Canada Act.

<https://ypfs.som.yale.edu/node/18406>

(BoC 2020s) Bank of Canada (BoC). 2020. "Provincial Money Market Purchase Program (PMMPP) Operations Data." Bank of Canada. Yale Program on Financial Stability Resource Library.

Spreadsheet published by the BoC reporting PMMPP operations data.

<https://ypfs.som.yale.edu/node/18496>

(BoC 2020t) Bank of Canada (BoC). 2020. "Provincial Money Market Purchase Program (PMMPP) Transaction Data." Bank of Canada. Yale Program on Financial Stability Resource Library.

Spreadsheet published by the BoC reporting PMMPP transaction-level data.

<https://ypfs.som.yale.edu/node/18495>

(BoC 2020u) Bank of Canada (BoC). April 2020. "Monetary Policy Report—April 2020." Bank of Canada.

Quarterly report of the Bank of Canada's Governing Council presenting the Bank's base-case projection for inflation and growth in the Canadian economy, and its assessment of risks.

<https://ypfs.som.yale.edu/node/18289>

(BoC 2020v) Bank of Canada (BoC). May 14, 2020. "Financial System Review—2020." Financial System Review.

Bank of Canada annual study analyzing the impact of COVID-19 on the Canadian financial system.

<https://ypfs.som.yale.edu/node/18264>

(BoC 2020w) Bank of Canada (BoC). May 21, 2020. "Quarterly Financial Report—First Quarter 2020."

Bank of Canada quarterly report for the period ended March 31, 2020.

<https://ypfs.som.yale.edu/node/18192>

(BoC 2020x) Bank of Canada (BoC). July 2020. "Monetary Policy Report—July 2020." Bank of Canada.

Quarterly report of the Bank of Canada's Governing Council presenting the Bank's base-case projection for inflation and growth in the Canadian economy, and its assessment of risks.

<https://ypfs.som.yale.edu/node/18290>

(BoC 2020y) Bank of Canada (BoC). August 28, 2020. "Quarterly Financial Report—Second Quarter 2020."

Bank of Canada quarterly report describing financial conditions for the period ending June 30, 2020.

<https://ypfs.som.yale.edu/node/18193>

(BoC 2020z) Bank of Canada (BoC). October 2020. "Monetary Policy Report—October 2020." Bank of Canada. Yale Program on Financial Stability Resource Library.

BoC publication reviewing monetary policy operations and programs for the period of July 2020 to September 2020.

<https://ypfs.som.yale.edu/node/18291>

(BoC 2020aa) Bank of Canada (BoC). November 9, 2020. "Quarterly Financial Report—Third Quarter 2020."

BoC quarterly report describing financial conditions for the period ending September 30, 2020.

<https://ypfs.som.yale.edu/node/18194>

(BoC 2021e) Bank of Canada (BoC). April 2021. "Monetary Policy Report—April 2021." Yale Program on Financial Stability Resource Library.

Quarterly report of the BoC's Governing Council presenting the Bank's base-case projection for inflation and growth in the Canadian economy, and its assessment of risks.

<https://ypfs.som.yale.edu/node/18293>

(BoC 2021f) Bank of Canada (BoC). April 21, 2021. "Annual Report 2020." Yale Program on Financial Stability Resource Library.

BoC annual report detailing the Bank's operations and financial statements.

<https://ypfs.som.yale.edu/node/18195>

(BoC 2021g) Bank of Canada (BoC). July 2021. "Monetary Policy Report—July 2021." Bank of Canada. Yale Program on Financial Stability Resource Library.

Quarterly report of the BoC's Governing Council presenting the Bank's base-case projection for inflation and growth in the Canadian economy, and its assessment of risks.

<https://ypfs.som.yale.edu/node/18294>

(Bulusu and Gungor 2017) Bulusu, Narayan, and Sermin Gungor. 2017. "The Life Cycle of Government of Canada Bonds in Core Funding Markets." Bank of Canada/Central Bank of Canada/La Banque du Canada. Yale Program on Financial Stability Resource Library.

BoC Review article analyzing the life cycle of Government of Canada (GoC) bonds in the context of central bank support for core funding markets.

<https://ypfs.som.yale.edu/node/18394>

(Fontaine, Garriott, Johal, Lee, and Uthemann 2021) Fontaine, Jean-Sébastien, Corey Garriott, Jesse Johal, Jessica Lee, and Andreas Uthemann. February 24, 2021. "COVID-19 Crisis: Lessons Learned for Future Policy Research." 2021-2.

Bank of Canada Staff Discussion Paper reviewing the initial impacts of the COVID-19 crisis on Canadian fixed-income markets.

<https://ypfs.som.yale.edu/node/18278>

(Gravelle 2021) Gravelle, Toni. March 23, 2021. "Market Stress Relief: The Role of the Bank of Canada's Balance Sheet." Speech presented at the CFA Society Toronto, Toronto, Ontario, March 23.

Bank of Canada Deputy Governor Toni Gravelle's speech regarding the central bank's actions during the COVID-19 economic downturn.

<https://ypfs.som.yale.edu/node/18168>

(Gravelle and Wilkins 2021) Gravelle, Toni, and Carolyn A. Wilkins. 2021. "The Bank of Canada's Response in 2020 to the Covid-19 Pandemic." In *Monetary Policy and Central Banking in the Covid Era*, 61-79. Edited by Bill English, Kristin Forbes, and Angel Ubide. London: CEPR Press.

Book chapter focusing on the actions taken by the Bank of Canada to address the economic and financial market fallout from the pandemic.

<https://ypfs.som.yale.edu/node/18296>

(PBO 2020) Office of the Parliamentary Budget Officer (PBO). June 25, 2020. "Provincial Government Liquidity Outlook." Office of the Parliamentary Budget Officer. Yale Program on Financial Stability Resource Library.

Report analyzing BoC support for provincial governments in response to the impacts of COVID-19.

<https://ypfs.som.yale.edu/node/18392>

(Solovieva and Hostland 2020) Solovieva, Maria, and Doug Hostland. April 21, 2020. "Bank of Canada Goes All-In to Safeguard Financial Stability." TD Economics. Yale Program on Financial Stability Resource Library.

Market notice describing stabilization measures implemented by the BoC in response to COVID-19.

<https://ypfs.som.yale.edu/node/18259>

Copyright 2022 © Yale University. All rights reserved. To order copies of this material or to receive permission to reprint any or all of this document, please contact the Yale Program on Financial Stability at ypfs@yale.edu .
